Homework 3

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Problem and gap

Economic complexity research shows that countries' development depends on the diversity of productive capabilities embedded in their economies. Hidalgo and Hausmann (2009) treat international trade as a bipartite country-product network and show that network-based complexity indicators are strongly correlated with income per capita and predict future growth, since development builds on existing capabilities in a path-dependent way. A key question is how countries acquire new capabilities. Bahar and Rapoport (2018) argue that tacit knowledge diffuses only through direct human interaction, so knowledge flows follow migration routes. Using bilateral migration, trade, and FDI data, they find that migration is significantly associated with new product exports, and that this effect depends strongly on migrants' educational attainment. Bahar et al. (2020) extend this to innovation: migrant inventors import the technological knowledge of their home countries, helping host countries enter new technological fields and gain technological advantage.

These studies treat migration as a set of independent country pairs. Yet the surrounding network structure may shape how knowledge circulates. Schilling and Phelps (2007) argue that dense local clustering increases a network's information transmission capacity: multiple pathways improve the fidelity of information, and clusters foster trust, reciprocity norms, and a shared understanding, which supports the transfer of tacit knowledge. In a panel of 1,106 firms in 11 industry alliance networks, they find that clustering raises firm patenting when it is combined with high reach, i.e., short paths to a wide range of firms. Fleming et al. (2007), in contrast, find in regional inventor networks that local clustering alone does not improve innovation, whereas shorter path lengths and the connection of previously isolated components increase future patenting. The evidence on clustering is therefore mixed: it appears to matter, if at all, in combination with access to diverse knowledge sources rather than on its own.

However, this evidence comes from firm and inventor collaboration networks, not from country-level migration networks, and not from outcomes measured by economic complexity.

Gap statement: Existing research links bilateral migration to countries' export and technological structures, and shows that clustering and reach in collaboration networks affect innovation. To our knowledge, however, no study examines whether the local structure of the international migration network, in particular the prevalence of closed triads, is associated with subsequent changes in countries' economic complexity (ECI), once the skill composition of migrants is taken into account.

Research question and hypothesis

Research question: Are countries embedded in more closed triads of the bilateral migration network associated with larger subsequent increases in their ECI?

Hypothesis: Countries with a higher weighted, directed clustering coefficient in the migration network experience larger subsequent ECI increases than otherwise similar countries.

Mechanism: Following Bahar and Rapoport, migrants carry tacit, origin-specific knowledge that moves only through direct interaction. Following Schilling and Phelps, repeated and overlapping interactions in closed triads build trust and reciprocity and provide multiple routes for the same information, which improves the transfer of complex, tacit knowledge. Following Hidalgo and Hausmann, complexity grows through the accumulation of capabilities, so better knowledge transfer should translate into ECI growth. Fleming et al. show that clustering alone may not suffice for innovation, so the estimate tells us which mechanism dominates.

Data and Method

Nodes and links: Nodes are OECD countries, links are directed and weighted: j → i is the annual inflow of immigrants of nationality j into country i, normalized by the destination's population. We filter weak links with Disparity Filter (Serrano et al., 2009).

Data: Migration (OECD International Migration Database), ECI growth (OEC), controls: GDP per capita, population, education level (OECD).

Measure: Closed triads: the weighted, directed clustering coefficient (Fagiolo, 2007) captures the share of a country's possible triangles that are actually closed, i.e., the origin countries of its immigrants are themselves linked by migration.

Estimation: Panel regression with country and year fixed effects, country-clustered errors. Dependent variable: ECI change between t and t+k, explanatory variable: triangle measure at t. We interpret results as predictive, not causal.

Plan B: If flow data are patchy: OECD DIOC bilateral stocks. If variance is low: longer windows or a binary measure.

Expected Results and Impact

Expected results: If the hypothesis holds, countries with a higher weighted, directed clustering coefficient at t show larger ECI increases between t and t+k, controlling for GDP per capita, initial ECI, population and education. This would support the view that overlapping migration ties help tacit knowledge circulate (Schilling & Phelps 2007). A null or negative coefficient is also informative. In line with Fleming et al. (2007), it would suggest that clustering alone is not enough, and that reach or bridging ties matter more for knowledge flows between countries.

Impact: If closed triads predict ECI gains, the result would be useful for three groups. Policymakers in host countries and at the OECD could use the migration network structure as an additional factor when assessing which migration patterns are likely to support diversification, while recognizing that the finding is predictive rather than causal and that host countries cannot directly shape ties among origin countries. Development and regional planners could use triad measures as an early indicator of where migration is likely to coincide with diversification, since the measure is computed from public flow data. Researchers gain a country-level test of the closure mechanism, which has so far been shown mainly for firms and inventors (Schilling & Phelps 2007; Fleming et al. 2007). A null or negative result would also be informative: it would suggest that the composition of origins does not matter beyond skills and size, or that reach and bridging ties matter more.

References

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